Cambridge (CIE) IGCSE Maths
Paper 1: Non-Calculator (Set F)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 30 minutes)

Total marks

/ 80

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- Calculators must not be used in this paper.

Materials

- List of formulas

Scan here to mark your mock exam

or visit the mock exams landing page for this course

1 The attendance at a football match is 11 678.

i) Write 11 678 in words.

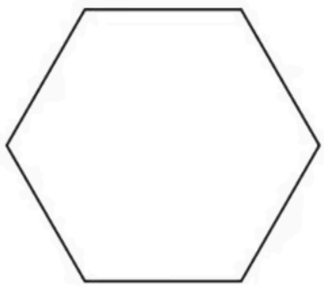
[1]

ii) Write 11 678 correct to the nearest 100.

[1]

(2 marks)

2 The diagram shows a regular polygon.



i) Write down the mathematical name for this shape.

[1]

ii) Write down the order of rotational symmetry of this shape.

[1]

(2 marks)

3 Using numbers from 55 to 85, write down

i) a multiple of 23,

[1]

ii) a factor of 120,

[1]

iii) a common multiple of 8 and 12,

[1]

iv) a number that is **both** square **and** odd,

[1]

v) a number that has exactly 2 factors.

[1]

(5 marks)

4

i) Write down the fraction of the rectangle that is shaded.

[1]

ii) Find the percentage of the rectangle that is **not** shaded.

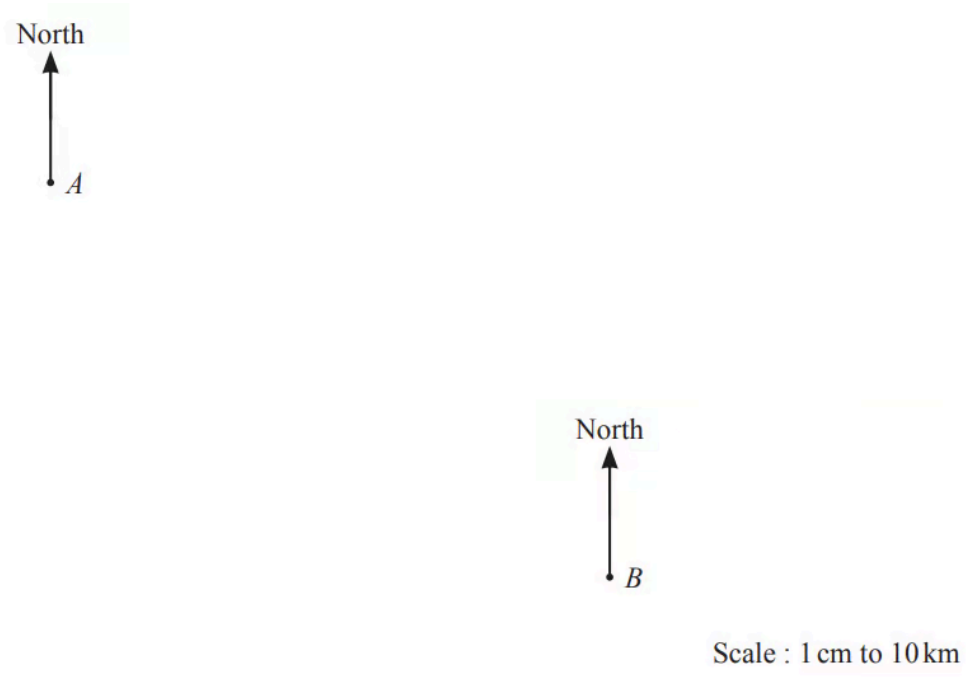
[1]

(2 marks)

5 Write $\frac{60}{105}$ in its simplest form.

(1 mark)

6 The scale drawing shows the positions of two towns, A and B.
The scale is 1 centimetre represents 10 kilometres.



i) Work out the actual distance between town A and town B.

.... km [2]

ii) Town C is 85 km from town A on a bearing of 100°.

On the scale drawing, mark the position of town C.

[2]

(4 marks)

7 Nora makes a birthday cake.

Nora has a packet containing 250g of cherries.

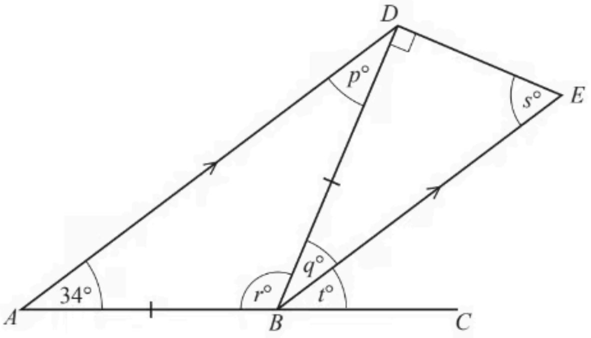
She uses $\frac{7}{10}$ of the cherries in the cake.

Find the mass of cherries she has left.

.... g

(2 marks)

8



NOT TO SCALE

In the diagram, ABC is a straight line.
AD is parallel to BE, angle BAD = 34° and $AB = BD$.

i) Complete the statements.

a) p = ... because ....

[2]

b) q = ... because ....

[2]

ii) Work out the value of r and the value of s.

r = ...

s = ... [2]

iii) Find the value of t and give a reason for your answer.

t = ... because ....

[2]

(8 marks)

9 Calculate $7 \times (2 \times 4) + 8 \times (2 + 6)$

(2 marks)

10 The temperature on Monday was -7°C.
The temperature on Tuesday was 5°C lower than on Monday.
The temperature on Wednesday was 8°C higher than on Tuesday.

Find the temperature on Wednesday.

.... °C

(2 marks)

11 Factorise $5p + pt$.

(1 mark)

12 Solve these equations.

$$12x - 3 = 4x + 21$$

$x =$ ....[2]

(2 marks)

13 Peter works these hours each week at a café.

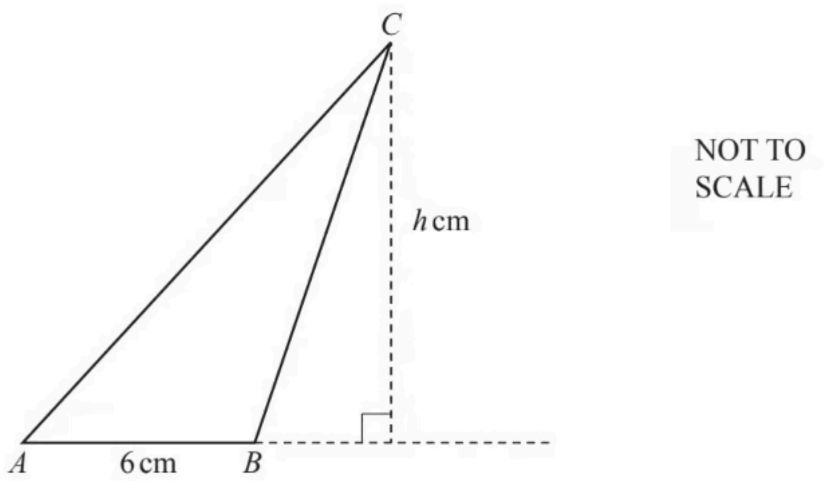
Day	Time
Monday	0830 to 16 00
Tuesday	1000 to 17 00
Thursday	0830 to 16 30
Saturday	0800 to 18 30

Work out the number of hours he works in one week.

.... hours

(2 marks)

14



The area of triangle ABC is 27 cm^2 and AB = 6 cm.

Calculate the value of h.

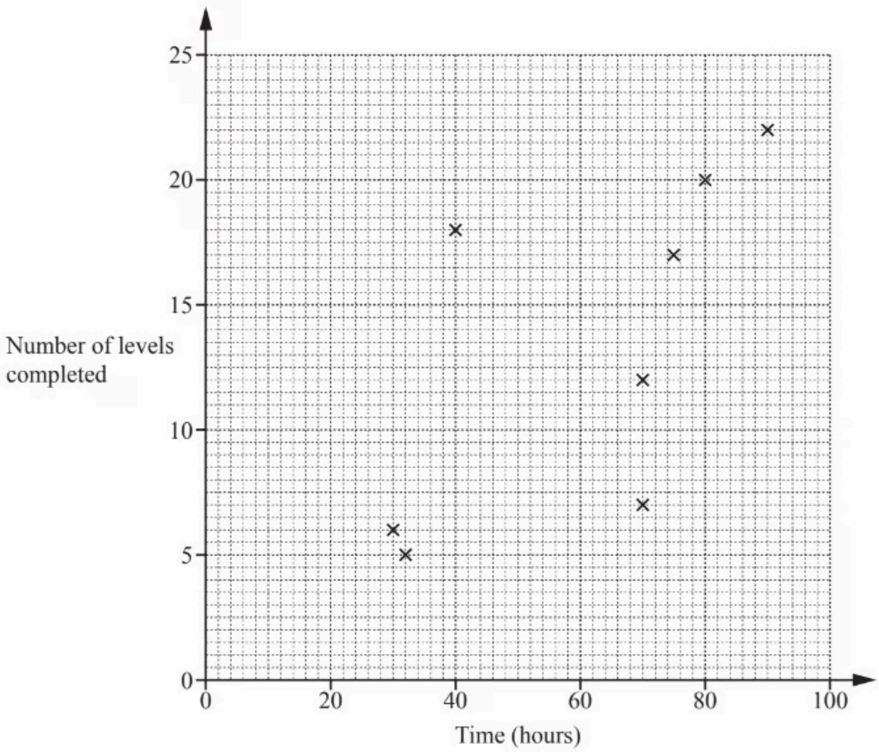
h = ...

(2 marks)

15 (a) Lucy asked 12 people how many hours they each spent playing a computer game and the number of levels they each completed in one month.

The results are shown in the table.

Time spent playing (hours)	90	32	70	75	30	70	40	80	40	65	50	32
Number of levels completed	22	5	12	17	6	7	18	20	8	15	11	9



Complete the scatter diagram.
The first eight points have been plotted for you.

[2]

(2 marks)

(b) One person completes more levels per hour than any of the others.

On the scatter diagram, put a ring around the point for this person.

[1]

(1 mark)

(c) What type of correlation does this scatter diagram show?

[1]

(1 mark)

(d) On the scatter diagram, draw a line of best fit.

[1]

(1 mark)

(e) Another person, Monika, completed 19 levels but forgot to record the time spent playing.

Use your line of best fit to estimate the number of hours that Monika spent playing.

.... hours [1]

(1 mark)

16 Emma records the number of letters in each word in a sentence.

7	1	5	4	2	4	5	3	3	1	2	4

Find

i) the median,

[2]

ii) the mode,

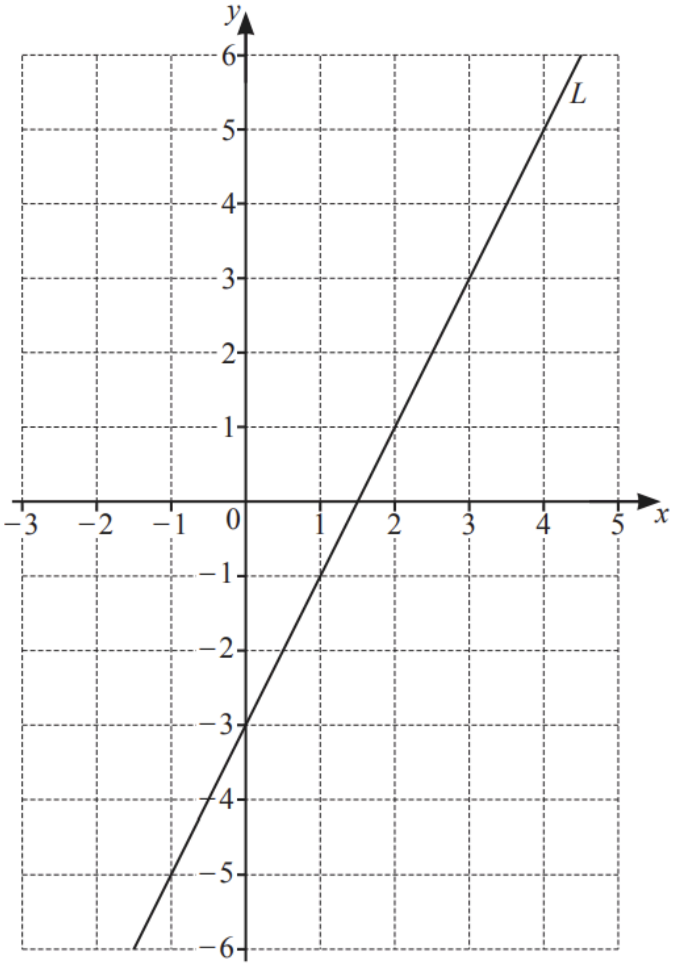
[1]

iii) the range.

[1]

(4 marks)

17



Find the equation of line L in the form $y = mx + c$

y = ...

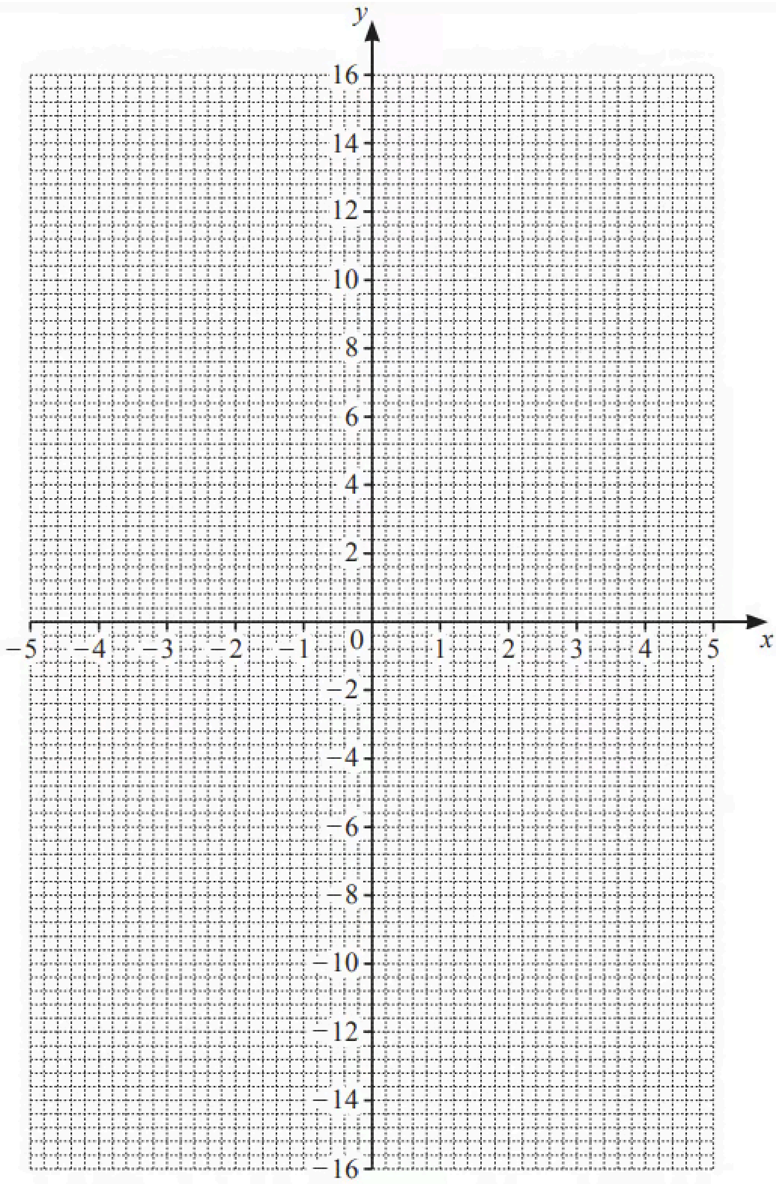
(2 marks)

18 (a) Complete the table of values for $y = \dfrac{15}{x}$

x	-5	-3	-2	-1		1	2	3	5
y				-15		15			

(3 marks)

(b) On the grid, draw the graph of $y = \dfrac{15}{x}$ for $-5 \leqslant x \leqslant -1$ and $1 \leqslant x \leqslant 5$.



(4 marks)

(c) On the grid, draw the line $y = 6$.

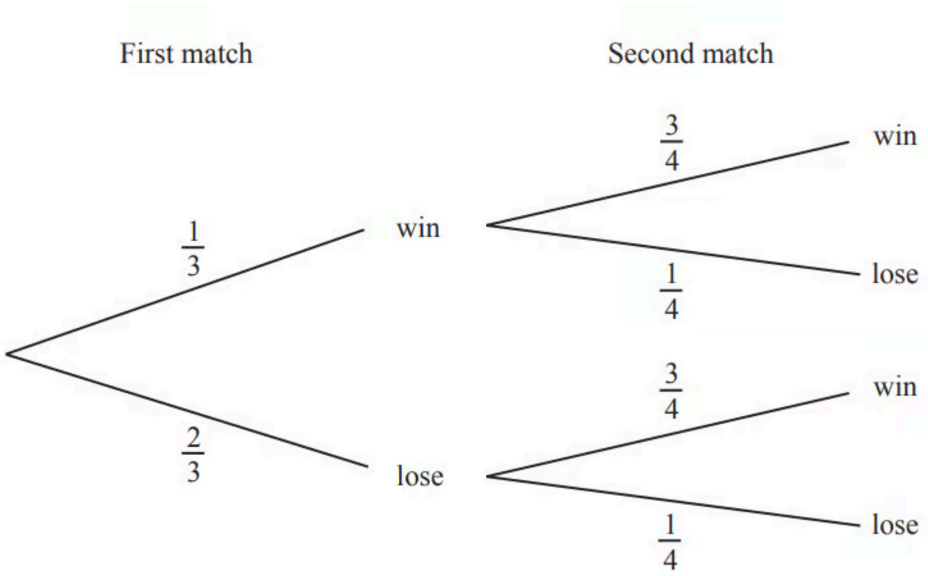
(1 mark)

(d) Use your graph to solve $\frac{15}{x} = 6$.

$x =$ ....

(1 mark)

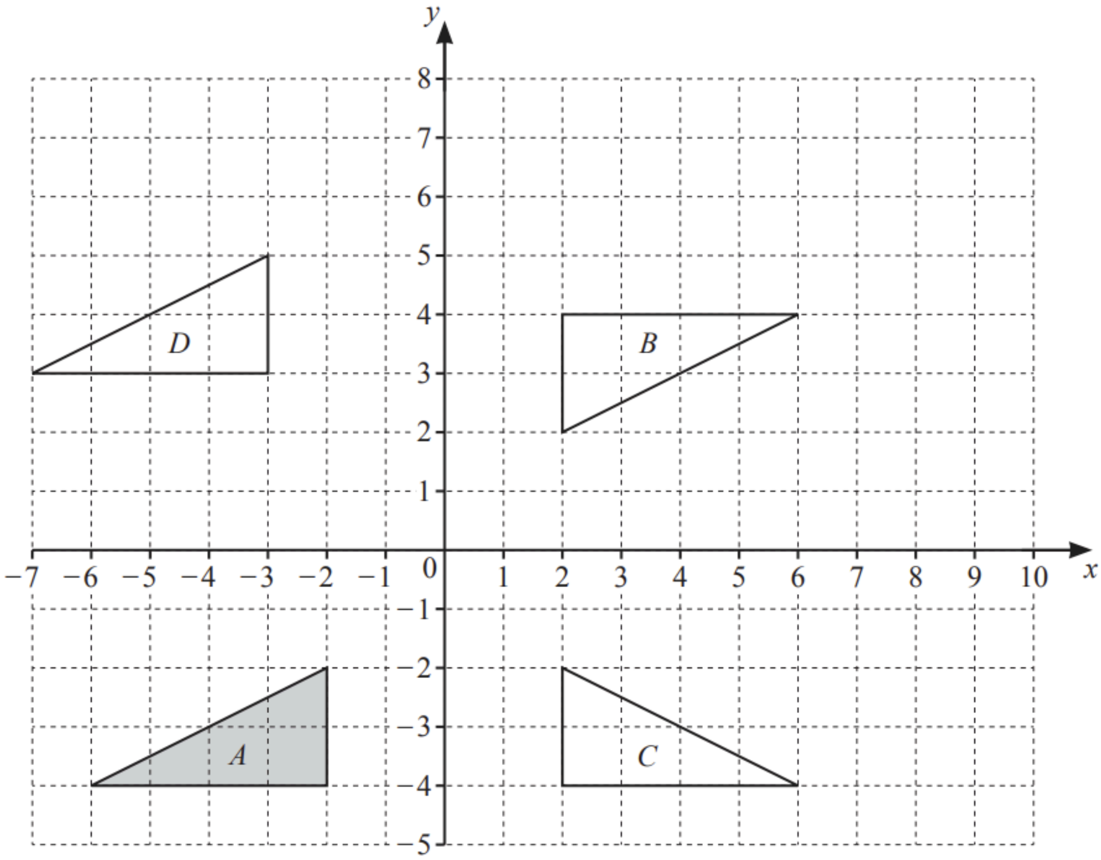
19 A soccer team plays two matches.
The tree diagram shows the probability of the team winning or losing the matches.



Find the probability that the soccer team wins at least one of the two matches.

(3 marks)

20 (a)



Describe fully the **single** transformation that maps

i) triangle A onto triangle B,

[3]

ii) triangle A onto triangle C,

[2]

iii) triangle A onto triangle D.

[2]

(7 marks)

(b) On the grid, enlarge triangle A by scale factor 0.5, centre (4, 0).

(2 marks)

21 Make p the subject of this formula.

$$H = 7p - 3$$

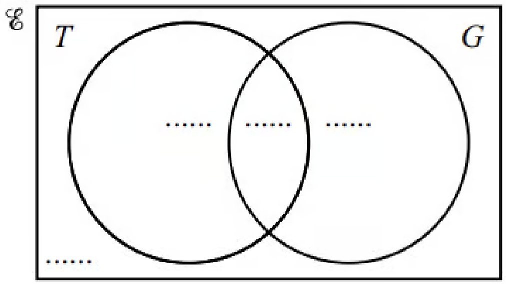
$p =$ ....

(2 marks)

22 (a) $\mathscr{E}$ = {children who go to the park}
T = {children who play tennis}
G = {children who play golf}

120 children go to the park.
50 play tennis.
75 play golf.
25 do not play tennis or golf.

Complete the Venn diagram.



[2]

(2 marks)

(b) Find n($T \cap G$).

(1 mark)

23 Solve the simultaneous equations.

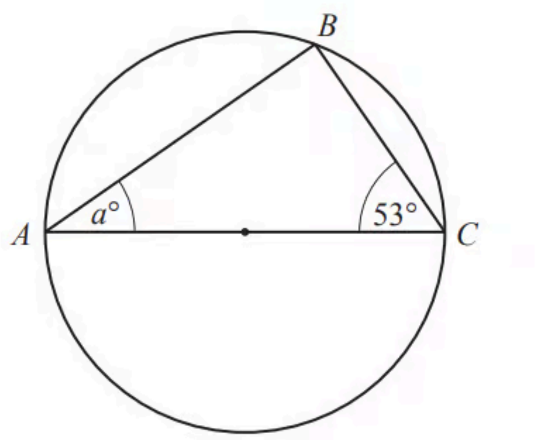
You must show all your working.

$$3x - 8y = 22$$
$$x + 4y = 4$$

x = ...
y = ...

(3 marks)

24



A, B and C are points on a circle.
AC is a diameter of the circle.

Find the value of a.

a = ...

(2 marks)